ChatGPT

No-Vendor Roadmap for Testing Crypto Perpetual Alpha Hypotheses

Decision rule

The practical line is not “can I imagine a signal,” but “can I test it with public data, a conservative all-taker execution model, and no hidden dependence on unavailable microstructure.” Under that standard, families that can survive one-bar-late entries, close-confirmation triggers, and public OHLCV plus mark/funding/OI are the right place to start. Families that only make sense if you can see trade aggressor flow, queue position, or deeper-book exhaustion need to be rewritten into slower, confirmation-based versions or dropped. That is especially true for small aggressive accounts, where paying taker fees is realistic but assuming passive queue priority is not.

One family in the list has unusually strong support from standard bar data alone: time-series momentum and its volatility-managed versions. The classic TSMOM construction is return-history based, and volatility management is built from realized variance scaling rather than order-book features, so these ideas are naturally Tier 1 if you can build a funding-adjusted return series for perps. [1]

What public venue data actually covers

Kraken’s official derivatives stack is stronger than a plain “bars only” venue. Its public futures ticker endpoints expose mark price, index price, funding rate and open interest; its candles API supports trade, mark, and spot tick types; its historical funding endpoint is public; and its analytics endpoint exposes time-bucketed series for open-interest, liquidation-volume, CVD, spreads, liquidity, slippage, future-basis, funding, and more. Kraken also exposes public order books, a recent trade-history endpoint that is explicitly limited to the past seven days or the most recent engine restart, and separate paginated public execution, order, and mark-price event-history endpoints. That combination makes Kraken viable not just for Tier 1 work, but for a useful Tier 2/Tier 3 substitute on selected problems. [2]

Broader venues widen the menu. Bybit’s public docs expose kline, mark-price kline, index-price kline, funding-rate history, open interest, 1000-level order-book snapshots, recent public trades, and instrument metadata including `launchTime`, status, `fundingInterval`, and pre-launch structures. Its public API documentation also points to downloadable public OHLCV and trade-history datasets, and the recent-trades documentation explicitly notes archived historical trades can be downloaded from the website. OKX exposes instruments with `state` and `listTime`, funding-rate history, open interest, mark price plus mark-price candles/history, market trades/order books, and a public liquidation-orders channel; its official historical-data page advertises downloadable perpetual funding history from March 2022 onward and high-resolution L2 order-book data from March 2023 onward. Coinbase International is more mixed: it exposes candles, historical funding, public market-data WebSockets, and LEVEL1/LEVEL2 depth feeds, and its instruments endpoint includes current `open_interest`, `funding_interval`, and quote fields such as index price, mark price, and predicted funding, but the documentation reviewed here does not expose a dedicated historical open-interest endpoint comparable to Bybit or OKX. [3]

That gives a clean testing hierarchy. Tier 1 is enough for slower continuation, breakout, session, funding-window, and trend-following ideas. Tier 2 or forward capture becomes important when the edge depends on spread state, top-of-book stability, or shallow-depth behavior. Tier 3 matters when the current translation truly requires trade-print sequencing, event replay, or queue-aware execution. Where official downloads exist, they can partly replace a vendor; where they do not, the substitute is forward capture, not optimism. [4]

Family-by-family roadmap

In the table below, the “best no-vendor variant” column names both the cleanest implementation and the less depth-sensitive redesign. The “current status” column is carrying four things at once: the minimum must-have data, the confidence-boosting data that would help but may be missing, the condition that would make the current translation discardable, and the hypothesis that is still worth preserving.

family	best no-vendor variant	minimum data tier	Kraken feasibility	broad-venue feasibility	capture requirement	current status	recomme
liquid leader continuation	Trade BTC/ ETH/SOL-style leader continuation only after a full bar closes through the trigger with rising OI; redesign away from stop-on-touch and book-following entries	Tier 1	Yes; strong on major perps via candles, mark/ index, OI, funding, analytics	Yes	None	**Keep.** Req: bars + mark/ index + OI. Better: CVD/ aggressor flow and shallow-depth stability. Discard the current translation if the edge only exists on first-touch entries or assumed maker fills. Preserve the slower “impulse + OI confirmation + next-bar continuation” hypothesis.	test_now_

family	best no-vendor variant	minimum data tier	Kraken feasibility	broad-venue feasibility	capture requirement	current status	recomme
prior-high momentum	Break of prior day/week high only if the breakout bar closes above the level and OI expands; redesign away from wick breaks	Tier 1	Yes	Yes	None	**Keep.** Req: intraday bars, mark or trade candles, OI. Better: trade prints for follow-through quality. Discard if the alpha disappears once “touch of prior high” becomes “close above prior high.” Preserve close-confirmed breakout momentum.	test_now_

family	best no-vendor variant	minimum data tier	Kraken feasibility	broad-venue feasibility	capture requirement	current status	recomme
volatility-managed TSMOM	Funding-adjusted TSMOM on daily or 4h perp returns, scaled by realized volatility; no microstructure dependence	Tier 1	Yes on liquid majors	Yes	None	**Keep.** Req: return history, funding history, realized vol. Better: longer cross-sectional history and delisted contracts. Discard only if the signal fails under funding-adjusted total returns and all-taker turnover assumptions. Preserve low-turnover trend plus vol-scaling.	test_now_
post-breakout retest	Enter only after a breakout, retest, and bar-close reclaim of the retest zone; redesign away from touch-the-level buys	Tier 1	Yes	Yes	Useful, not required	**Redesign.** Req: bars, level definition, optional OI. Better: L2 absorption and trade prints around the retest. Discard the current translation if it needs exact retest fills inside the bar. Preserve the “retest survives and reclaims on close” setup.	redesign_

family	best no-vendor variant	minimum data tier	Kraken feasibility	broad-venue feasibility	capture requirement	current status	recomme
sector/ theme ignition	Endogenous sector breadth ignition: trade laggards only when a mapped theme leader breaks and multiple same-theme names confirm with OI breadth; redesign away from narrative-only chasing	Tier 1	Weak on Kraken because breadth is limited	Yes; best on Bybit/OKX, partial on Coinbase	Helpful for future breadth expansion	**Hypothesis only.** Req: sector map + multi-asset bars + OI. Better: broader listing universe and reliable theme/event labels. Discard the current translation if it depends on discretionary theme tagging or a tiny venue universe. Preserve the sector-breadth contagion hypothesis, not the loose narrative wrapper.	preserve_

family	best no-vendor variant	minimum data tier	Kraken feasibility	broad-venue feasibility	capture requirement	current status	recomme
post-catalyst base	Build a dated catalyst ledger, then test base formation and range-hold after the catalyst impulse; redesign away from discretionary “story still good” entries	Tier 1	Partial; feasible but sample is thin	Yes	Event ledger is required	**Needs structure first.** Req: event timestamps + bars + OI/ funding. Better: venue-specific trades/L2 around the event. Discard the current translation if catalysts are not timestamped ex ante. Preserve the post-event base-and-break pattern.	build_eve
failed catalyst short	Dated catalyst ledger plus failure rule: close back into pre-event range or lose the post-event base with stalled OI; redesign away from narrative shorts	Tier 1	Partial	Yes	Event ledger is required	**Needs structure first.** Req: event timestamps, bars, OI. Better: prints and depth around the failure. Discard the current translation if “catalyst failure” cannot be defined mechanically. Preserve the post-event failure hypothesis.	build_eve

family	best no-vendor variant	minimum data tier	Kraken feasibility	broad-venue feasibility	capture requirement	current status	recomme
small-cap reversal	If kept, rewrite as “eligible listed-perp loser reversal after overshoot and stabilization,” entered on bar confirmation; avoid microcap knife-catching	Tier 1	Poor fit on Kraken	Partial to yes on Bybit/OKX; weaker on Coinbase for OI history	Forward capture is useful	**Current translation should go.** Req: broad small-cap perp universe, lifecycle metadata, bars. Better: deep book, prints, richer history. Discard the current version if it assumes tiny-cap perp breadth, intrabar reversal fills, or microstructure visibility you do not have. Preserve only a much slower post-washout loser-reversal idea.	discard_c

family	best no-vendor variant	minimum data tier	Kraken feasibility	broad-venue feasibility	capture requirement	current status	recomme
liquidation/ deleveraging reclaim	Reclaim after forced-flow washout, defined by liquidation spike and/or sharp OI drop plus close back inside prior range; redesign away from tape-reading the bottom	Tier 2	Yes; Kraken analytics materially help	Yes; strongest on OKX, workable on Kraken, weaker elsewhere without capture	Forward capture is useful and often the substitute for missing history	**Keep, but only in a mechanical form.** Req: bars + OI + liquidation analytics/ channel substitute. Better: full liquidation tape and trades. Discard if the current translation depends on exact microsecond capitulation sequencing. Preserve the “forced-flow flush then reclaim” hypothesis.	run_forwa

family	best no-vendor variant	minimum data tier	Kraken feasibility	broad-venue feasibility	capture requirement	current status	recomme
listing/perp-launch short or fade	Fade only after price-discovery failure: first failed reclaim or close-back-inside after launch; redesign away from unconditional day-one shorting	Tier 1	Partial to weak; launch metadata are not the venue's strength	Yes; best on Bybit/OKX via `launchTime` / `listTime` / prelaunch state	Required for future events and robustness	**Keep as a conditional setup, not a blanket rule.** Req: lifecycle metadata + first-bar history. Better: launch auction/depth context and repeated forward sample. Discard the current translation if it is just "short every launch." Preserve the "bad price discovery after launch" hypothesis.	run_forwa

family	best no-vendor variant	minimum data tier	Kraken feasibility	broad-venue feasibility	capture requirement	current status	recomme
funding-window	Trade only extreme-funding windows, either as short-horizon mean reversion or as an avoidance/filter rule; redesign away from every-window trading	Tier 1	Yes	Yes	None	**Keep.** Req: funding history/schedule, mark/index, optional OI. Better: premium and shallow-depth context near settlement. Discard if net edge vanishes once taker fees and realistic timing around settlement are imposed. Preserve extreme-window conditioning, not generic funding-window overtrading.	test_now_

family	best no-vendor variant	minimum data tier	Kraken feasibility	broad-venue feasibility	capture requirement	current status	recomme
ORB/session	Define synthetic crypto sessions mechanically and test opening-range breakouts with close confirmation; redesign away from arbitrary hand-picked sessions	Tier 1	Yes	Yes	None	**Keep.** Req: intraday bars and session clock. Better: top-of-book/ spread state at the break. Discard if the setup only works with ex post session choices or intrabar touch entries. Preserve a few fixed global-session definitions and test them honestly.	test_now_
cross-coin lead/lag	Slower lead/ lag only: 5–30 minute leader impulse followed by laggard underreaction and OI filter; redesign away from sub-second causal claims	Tier 1	Yes on major clusters	Yes	Useful if you later push to faster horizons	**Redesign.** Req: synchronized bars across coins, optional OI. Better: trades, quote stamps, and event replay. Discard the current translation if it relies on tick-level precedence or queue effects. Preserve bar-horizon underreaction across linked coins.	redesign_

family	best no-vendor variant	minimum data tier	Kraken feasibility	broad-venue feasibility	capture requirement	current status	recomme
weak-asset spike fade	Fade only when a structurally weak asset spikes into resistance, stretched funding/OI, and then closes back below the level; redesign away from first-wick fading	Tier 1	Partial; breadth is thinner	Yes; best on Bybit/OKX	Useful	**Redesign.** Req: bars, relative-strength ranking, optional OI/funding. Better: prints and shallow-depth exhaustion. Discard the current translation if it needs intrabar top-picking. Preserve "weak asset fails after spike" on bar confirmation.	redesign_
parabolic blowoff short	Multi-bar convex extension plus extreme crowding, then short only after a reversal/failed continuation close; redesign away from picking the exact top	Tier 1	Partial; sample count is lower	Yes	Useful because events are rare	**Keep only in delayed form.** Req: bars, OI/funding stretch. Better: liquidation flow and trades. Discard the current translation if it is a naked top-call without failure confirmation. Preserve the blowoff-then-failure hypothesis.	run_forwa

family	best no-vendor variant	minimum data tier	Kraken feasibility	broad-venue feasibility	capture requirement	current status	recomme
failed breakout short	Short only after break-back-below the trigger and loss of the breakout bar low; no need for depth if entries are confirmation-based	Tier 1	Yes	Yes	None	**Keep.** Req: bars and level logic. Better: L2 confirmation that the breakout did not stick. Discard only if the signal vanishes once you require a confirmed failure rather than a wick rejection. Preserve mechanically defined failed-breakout shorts.	test_now_
generic shock reversal	Rewrite as an endogenous shock rule: n-sigma move plus OI/liquidation dislocation, then reversal back inside range; abandon the vague “shock” framing	Tier 1	Partial; Kraken analytics help	Yes	Useful for event tagging	**Current generic translation is not good enough.** Req: bars, volatility-normalized move, OI and ideally liquidation context. Better: event tags, trades, depth. Discard the current translation if “shock” is not defined in the data. Preserve only a narrower conditioned reversal rule.	discard_c

family	best no-vendor variant	minimum data tier	Kraken feasibility	broad-venue feasibility	capture requirement	current status	recomme
basis/ funding carry	Test as a sparse extreme-funding/basis overlay with full taker-cost assumptions, not as a generic standalone carry engine	Tier 1	Data-feasible	Data-feasible; best on Bybit/ OKX for richer history	None	**Standalone current translation should be discarded.** Req: funding, mark/index, basis proxy, execution-cost model. Better: hedge legs, borrow/spot financing, richer basis history. Discard if the strategy only works before taker costs and hedge frictions. Preserve extreme-funding carry as a filter or occasional overlay, not a core alpha sleeve.	discard_c

family	best no-vendor variant	minimum data tier	Kraken feasibility	broad-venue feasibility	capture requirement	current status	recomme
volatility breakout/ compression	Compression breakout on low realized-vol / inside-bar structure with OI expansion and close confirmation; no depth needed if entries are delayed	Tier 1	Yes	Yes	None	**Keep.** Req: bars, realized-vol or range-compression metric, optional OI. Better: spread/ liquidity state at breakout. Discard only if the edge depends on pre-break order-book pressure that you cannot test. Preserve close-confirmed compression breakouts.	test_now_

What to test first

The first batch should be the families that are naturally Tier 1, survive all-taker assumptions, and have enough repetition to produce a useful sample before you build anything fancy. On that basis, the clean opening slate is liquid leader continuation, prior-high momentum, volatility-managed TSMOM, funding-window, ORB/session, failed breakout short, and volatility breakout/compression. Those ideas can be expressed cleanly with public bars, mark/index, funding, and OI, and Kraken alone is enough to start building the machinery. Kraken's analytics also make liquidation/deleveraging reclaim more plausible than it would otherwise be, but that one still benefits from parallel forward capture because forced-flow signals are event-like rather than smooth. [5]

The second batch is the redesign batch: post-breakout retest, cross-coin lead/lag, weak-asset spike fade, and parabolic blowoff short. None of these should be tested in their reflexive trader-twitter form. They become testable only after you remove wick-touch entries, sub-second causality claims, and invisible L2 "feel." If the edge does not survive that rewrite, it was never really testable under your constraints.

The third batch is conditional on new scaffolding, not on better imagination. Post-catalyst base and failed catalyst short need a dated event ledger before they deserve a backtest. Listing/perp-launch fade and liquidation/deleveraging reclaim need forward capture because sample size, launch mechanics, and forced-flow episodes are too path-dependent to trust a thin retrospective reconstruction. Sector/theme ignition and small-cap reversal are not first-Kraken strategies; they are broad-universe strategies, and if you force them into a sparse venue they will tell you more about universe choice than about alpha.

Hard limits that should force redesign or discard

A translation should be discarded quickly if it depends on one of four hidden assumptions. The first is invisible execution quality: if the edge only works when you are filled on the touch, ahead in queue, or passively earning spread, it is not a no-vendor small-account strategy. The second is invisible state: if the alpha requires aggressor flow, full-depth exhaustion, or precise liquidation sequencing that you cannot reconstruct from public history, then the bar-based translation is not a faithful test. The third is undefined conditioning: “shock,” “theme,” and “catalyst” are not variables until they have timestamps and rules. The fourth is false economics: basis/funding carry often looks good before friction, but small aggressive all-taker accounts do not get to live in the pre-friction world.

The broad pattern is straightforward. Trend, continuation, breakout, compression, and carefully filtered funding-window ideas are testable now. Event families are testable only after you build a ledger. Microstructure-flavored reversals are testable only after they are slowed down and converted into confirmation-based bar rules. Standalone carry and vague shock-reversal translations deserve much less patience. The useful tension is here: Kraken’s public stack is rich enough to start honestly, especially because its analytics and event-history endpoints are stronger than many people expect, but the moment a hypothesis really needs breadth, launch metadata, or liquidation tape at scale, the roadmap bends away from “more clever backtest” and toward “broader venue plus forward capture.” [6]

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[1] Time series momentum - NYU Stern
https://w4.stern.nyu.edu/facdir/lpederse/papers/TimeSeriesMomentum.pdf?utm_source=chatgpt.com

[2] [5] Get tickers
https://docs.kraken.com/api-reference/market-data/get-tickers?utm_source=chatgpt.com

[3] Get Instruments Info | Bybit API Documentation
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[4] Export historical funding rates
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[6] Get trade history - Kraken Developers
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